Ikram Ul Haq

AI/ML Engineer · LLM Evaluation · Agents · Production Generative AI

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Profile

AI/ML engineer with 6+ years of production software and ML experience, 3+ of them shipping LLM and generative AI systems that people use. At Smartly I work on the generative pipelines behind customer ad content (text, images, video), own the production LLM gateway and LLM/CV observability, and built the evaluation pipeline for a non-deterministic AI agent. Previously AI Tech Lead at Nokia, leading the team that built a GenAI platform (RAG, agentic workflows, distributed LLM fine-tuning) on on-prem Kubernetes. I care about measuring what ships: defining what a correct result is, building the datasets that catch regressions, and instrumenting systems so failures arrive with enough context to fix.

Core Strengths

- **Evaluation as a deliverable:** agent and LLM eval pipelines, benchmark design on open datasets, LLM-as-judge, regression gating.
- **Shipping generative AI:** text, image and video generation pipelines in production; RAG; agentic workflows and tool calling (MCP).
- **Inference cost and latency:** model routing, usage and cost monitoring, tracing and latency diagnosis across distributed services; benchmark-driven model selection.
- **Model adaptation:** distributed fine-tuning of open LLMs on GPU clusters; comparing open models before committing to training.
- **Mentoring:** led and mentored an engineering team through code review, design review and sprint ownership.

Professional Experience

Machine Learning Engineer II

Smartly, Helsinki, Finland 03/2026 – Present

- Improve the generative AI and computer-vision pipelines that produce ad content (text, images, video) for Smartly's customers in production, at scale.
- Built a self-improvement evaluation pipeline for an internal AI agent (Hermes), automating quality assessment of non-deterministic agent outputs and feeding results back into iterative refinement.
- Evaluated LLM gateways and deployed LiteLLM (3 replicas) as the centralized production gateway for model routing, usage and cost monitoring, and key management.
- Led the LLM observability tooling assessment (Langfuse vs. MLflow) across 13 production applications (8 LLM, 5 CV/ML); rolled out tracing and monitoring with Langfuse and MLflow, with Evidently AI recommended for CV production monitoring.
- Contribute to MCP gateway standardization, giving agents shared, governed tool access across teams.

AI Tech Lead

Nokia, Espoo, Finland 03/2025 – 02/2026

- Owned the technical design and delivery of a modular GenAI platform spanning RAG, agentic workflows, and distributed LLM training on private, on-prem Kubernetes clusters.
- Led and mentored a team of engineers through code and design reviews; owned sprint goals and alignment with Nokia's internal AI strategy.
- Enabled distributed LLM training and fine-tuning across GPU nodes via Kubernetes-native scheduling.
- Architected observability-first microservices (Docker, Kubernetes, MLflow, OpenTelemetry, Jaeger, Prometheus, Grafana, Fluentd), giving end-to-end tracing across distributed request paths.

Machine Learning Engineer

Nokia, Espoo, Finland 05/2023 – 02/2025

- Designed and ran evaluations of open LLMs (Llama 3, Gemma, Mistral) for summarization, code generation, and chat on open benchmarks (CNN/DailyMail, CoNaLa, Chatbot Arena); findings informed model selection and a master's thesis on industrial LLM applicability.
- Developed on-prem, cloud-native microservices for LLM use cases, including RAG.

Machine Learning Engineer
Arbisoft, Lahore, Pakistan 08/2020 – 08/2022

- Built NLP tagging pipelines that classified and filtered user reviews for travel platforms (e.g., KAYAK), improving SEO and content discoverability.
- Implemented hotel-specific aspect extraction (cleanliness, service, etc.) for personalized search, and evaluated open-source machine-translation models for multilingual reviews.
- Delivered working prototypes directly to clients and communicated progress and trade-offs that shaped project scope.

Education

Master's in Intelligent Systems
University of the Balearic Islands, Palma, Spain 2023 – 2024

Thesis: Performance Analysis of Large Language Models on Summarization, Code Generation and Conversational AI.

Master's in Engineering of Data-intensive Intelligent Software Systems (EDISS, Erasmus Mundus)
Åbo Akademi University, Turku, Finland 2022 – 2024

Fully funded; one of 23 selected worldwide. Courses: Computer Vision, Cloud-Native Applications, Big Data Management, Data Science.

BS Computer Science
University of the Punjab, Lahore, Pakistan 2016 – 2020

Vice Chancellor's honors list. Courses: Algorithms, Data Structures, Systems Programming, Software Engineering.

Skills

- **Languages:** Python (primary), SQL, C++, Rust, Java, Kotlin
- **LLM & ML:** LiteLLM, Langfuse, vLLM, HuggingFace, LangChain, PyTorch, TensorFlow, scikit-learn
- **Platform & Ops:** Kubernetes, Docker, MLflow, OpenTelemetry, Jaeger, Prometheus, Grafana, Argo, Jenkins, SonarQube

Achievements, Certifications & Games

- **Bronze Medal**, ACM-ICPC Asia Regional; winner of 6+ university programming competitions (incl. NaSCon'19, NAMAL TechnoFest'19, PUCIT 2018); data science hackathon winner.
- Curated and published a large-scale AIS maritime dataset from Finnish waters (Åbo Akademi University).
- Deep Learning Specialization and TensorFlow Developer Certificate (DeepLearning.AI); IBM Advanced Data Science Specialization; SQL for Data Science; AWS Fundamentals (Cloud-Native, Serverless).
- **Mobile games:** Clash Royale player for 7+ years (10,653 trophies; player tag ulhaqi12).